

Day 9: Probability & Statistics (cont.)

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Revision

- Data modeling: Describe the data through model.
- For distribution modeling we can:
 - Use graphical method
 - Anderson – Darling statistic

Outlines

- I. Monte Carlo simulation
- II. Bootstrapping
- III. Instruction on Zeppelin Markdown & LaTeX

You take n samples.

n is so big, most will follow population distribution.





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1. Determine the distribution
2. Build a procedure
 - a) Roll the dice.
 - b) Note the results.
 - c) Do (a) again in n times.
3. Do it until the parameters are stable enough.

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#2. Predict without known distribution

Given the height of students in 2021 as below:

154, 163, 171, 165, 170, 175, 155, 160

Estimate the average height using Monte Carlo simulation.

Step 1. Discretize the variable

$$c = 1 + 3.322\log_{10}n \approx 4$$

$$r = \max - \min = 21$$

$$w = \frac{21}{4} \approx 5.25 \approx 5$$

154, 163, 171, 165, 170, 175, 155, 160

Height (X)	Frequency (count)
154 – 158	2
159 – 163	2
164 – 168	1
169 – 173	2
174 – 178	1

Step 2. Calculate P

154, 163, 171, 165, 170, 175, 155, 160

Height (X)	Frequency (count)	P
154 – 158	2	2/8
159 – 163	2	2/8
164 – 168	1	1/8
169 – 173	2	2/8
174 – 178	1	1/8

Step 3. Processing (normalization)

154, 163, 171, 165, 170, 175, 155, 160

Height (X)	Frequency (count)	P	Normalize
154 – 158	2	2/8	0 – 0.25
159 – 163	2	2/8	>0.25 – 0.5
164 – 168	1	1/8	>0.5 – 0.625
169 – 173	2	2/8	>0.625 – 0.875
174 – 178	1	1/8	>0.875 – 1

Step 4. Find $E[X]$

154, 163, 171, 165, 170, 175, 155, 160

Height (X)	Frequency (count)	P	Normalize	$E[X]$
154 – 158	2	2/8	0 – 0.25	156
159 – 163	2	2/8	>0.25 – 0.5	161
164 – 168	1	1/8	>0.5 – 0.625	166
169 – 173	2	2/8	>0.625 – 0.875	171
174 – 178	1	1/8	>0.875 – 1	176

Monte Carlo applications

1. Simulated data (resampling)
2. Estimate statistics
3. Find optimized n for statistical experiments

II. Bootstrapping

- It's another way of **resampling**.
- We want to estimate the sampling distribution of any statistic using random sampling methods.

154, 163, 171, 165, 170, 175, 155, 160

Step 1. Sampling from the data

- The sample is known as **bootstrapped dataset**.
- Sampling with replacement.
- The sample is the same size as the data.

154, 163, 171, 165, 170, 175, 155, 160

165, 155, 163, 165, 154, 175, 165, 160

Step 2. Calculating

165, 155, 163, 165, 154, 175, 165, 160

≈ 162.75

Step 3. Redo from step 1 for n trials

Assumptions

- The population is big, so if we pick number a (in X), then it's likely we will pick another one that is equal to a .
- The calculation in step 2 can be highly different each time running.

Comparisons

Monte – Carlo

- Output: A new simulated data, or an estimation.
- **Advantages:** Easy to implement.
- **Disadvantages:** Do not simulate statistics.

Bootstrapping

- Output: A new simulated data with distribution.
- **Advantages:** Easy to implement.
- **Disadvantages:** Can be inconsistent.

III. Zeppelin Markdown & LaTeX

- Zeppelin supports markdown using `%md`.
- The markdown also includes LaTeX rendering.

Markdown cheatsheets

- #: Heading 1.
- ##: Heading 2.
- ###: Heading 3.
- ****text****: Bold.
- **text**: Italic.
- > text: Quote.
- [title](link): Link embedding.
- ![alt text](image url): Image.

Note: Zeppelin markdown supports HTML rendering also!